

EECE 306 Lab 3 notebook. Point Charges, Superposition, and the Field Engine

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Fill in every TODO, run `publish('Lab03_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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Setup

```
clear; close all;
rng(306);
e0 = em.const.eps0();
```

Closed form comparison at ten radii

```
Q = 2e-9;
s = em.src.pointCharge(Q, [0 0 0]);
radii = (1:10)';
Enum = em.field.E(s, [radii zeros(10,2)]);
Eref = Q ./ (4*pi*e0*radii.^2);
relerr = abs(Enum(:,1) - Eref) ./ Eref;
disp('  r (m)    relative error')
disp([radii relerr])
```

r (m)	relative error
1.0000	0
2.0000	0
3.0000	0.0000
4.0000	0
5.0000	0.0000
6.0000	0.0000
7.0000	0.0000
8.0000	0
9.0000	0.0000
10.0000	0.0000

Symmetry on the bisector plane

Two equal charges on the x axis. On the plane $x = 0$ the x component must vanish.

```
q1 = em.src.pointCharge(1e-9, [-0.1 0 0]);
q2 = em.src.pointCharge(1e-9, [ 0.1 0 0]);
s2 = em.src.merge(q1, q2);
pts = [zeros(200,1) randn(200,2)];
Ev = em.field.E(s2, pts);
fprintf('max |Ex| on bisector      = %.3e V/m\n', max(abs(Ev(:,1)))));
fprintf('max |E| on the same plane = %.3e V/m\n', max(em.vec.mag(Ev)));
% The expected x component is zero, so a relative tolerance cannot be used.
% We chose an absolute threshold of 1e-6 times the maximum field magnitude
```

```
% on the plane. This threshold is tiny compared with the field itself while
% still allowing for floating-point cancellation between the two charges.
```

```
max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 6.631e+02 V/m
```

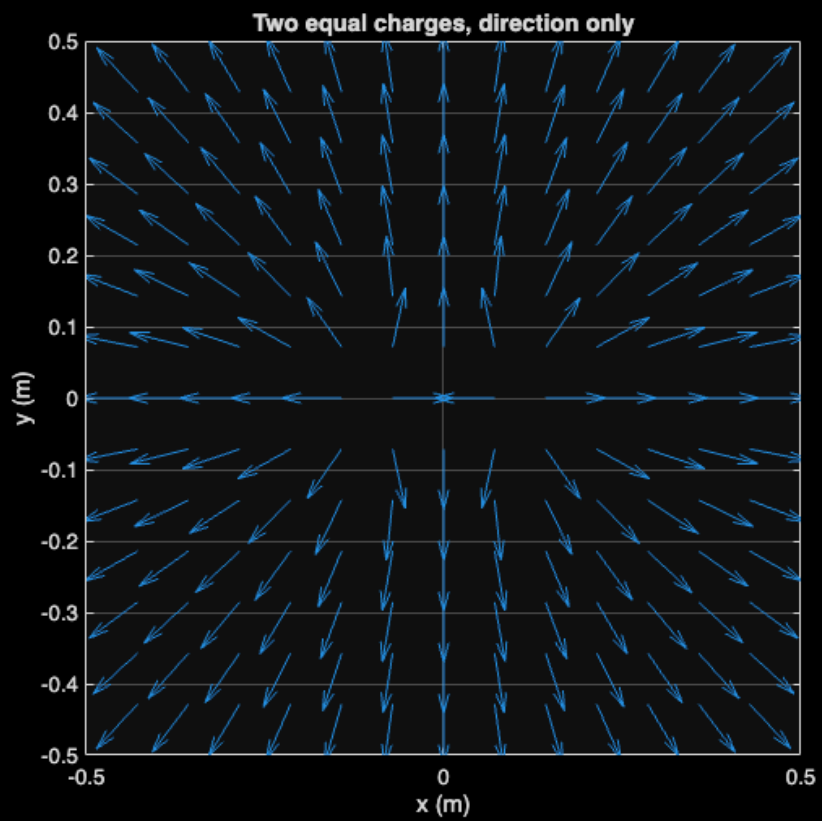
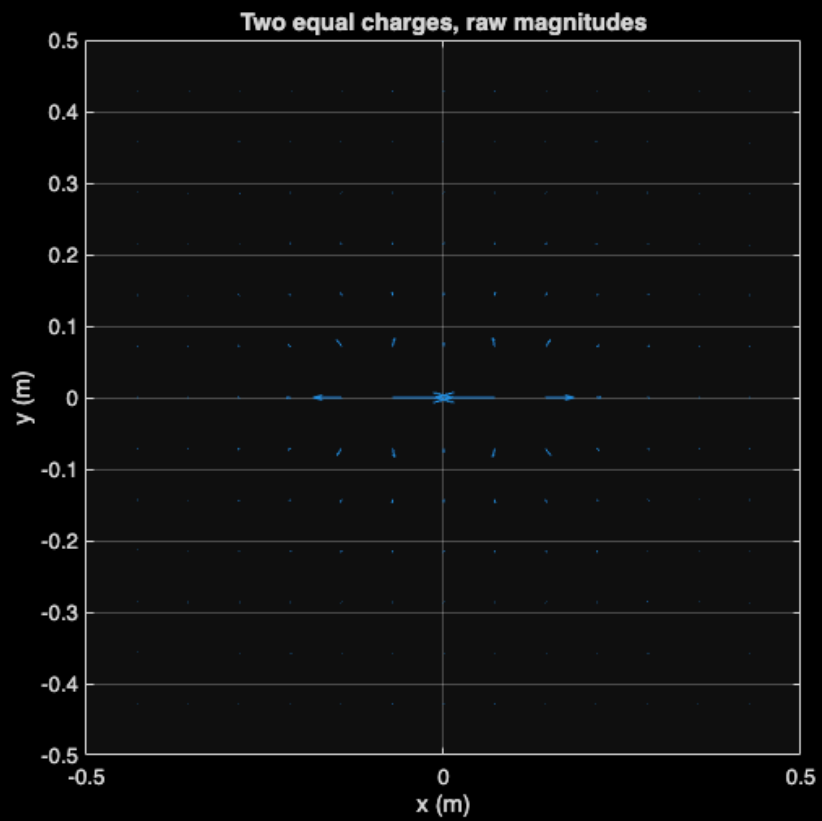
Dipole far field

```
sd = em.src.merge(em.src.pointCharge(1e-9,[0 0 0.05]), em.src.pointCharge(-1e-9,[0 0 -0.05]));
E1 = em.vec.mag(em.field.E(sd, [0 0 10]));
E2 = em.vec.mag(em.field.E(sd, [0 0 20]));
fprintf('|E(r)| / |E(2r)| = %.4f    (expect near 8 for a dipole)\n', E1/E2);
```

```
|E(r)| / |E(2r)| = 8.0003    (expect near 8 for a dipole)
```

The picture, raw and normalized

```
figure;
em.viz.quiver2(@(r) em.field.E(s2, r), [-0.5 0.5], [-0.5 0.5], 15, ...
    struct('normalize', false, 'title', 'Two equal charges, raw magnitudes'));
figure;
em.viz.quiver2(@(r) em.field.E(s2, r), [-0.5 0.5], [-0.5 0.5], 15, ...
    struct('normalize', true, 'title', 'Two equal charges, direction only'));
% The raw plot is difficult to read because the field strength changes by
% several orders of magnitude, so nearby arrows dominate the picture. The
% normalized plot clearly shows direction, but it gives up information about
% the relative field magnitude at each point.
```



```
pts5k = randn(5000,3) + 5;
tic; em.field.E(s2, pts5k); t = toc;
fprintf('em.field.E on 5000 points took %.3f s (requirement, under 2 s)\n', t);
```

```
em.field.E on 5000 points took 0.004 s (requirement, under 2 s)
```

Interpretation

The closed-form test checks the Coulomb constant, inverse-square magnitude, units, and radial components against a known analytical result. The symmetry test can reveal unequal treatment of mirrored elements or sign errors without needing a reference formula. The dipole scaling test checks whether superposition cancels the monopole term and produces the expected far-field $1/r^3$ behavior. Each test therefore catches a different class of error that the other two could miss.

Problems encountered

The supplied toolbox stored c0 without its required .m extension and did not include the cumulative Lab 2 test file. We restored those items, then corrected the Lab 3 implementation until the complete test suite passed.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
```

```
eta =
```

```
376.7303
```

```
test_lab01 checks complete
```

```
PASS test_lab01          0.05 s
```

```
test_lab02 checks complete
```

```
PASS test_lab02          0.03 s
```

```
5000-point field evaluation: 0.0007 s
```

```
test_lab03 checks complete
```

```
PASS test_lab03          0.20 s
```

```
3 passed, 0 failed, 3 total
```

```
All tests passing.
```